Comprehensive SQL Study Guide: Chapters 1, 2, and 3 (Intro)

1. Fundamentals of the SELECT Statement

The SELECT statement is the primary data retrieval tool in Oracle SQL. As a developer, you must master the syntax to display specific data attributes from database objects.**Core Syntax and Table Context** The basic execution structure follows a mandatory sequence: SELECT what_to_display FROM table_name;The primary tables utilized in this curriculum include:

- **EMPLOYEES (E):** Staffing details (ID, Names, Salary, Manager, Department).
- **DEPARTMENTS (D):** Organizational units.
- **LOCATIONS (L):** Physical addresses and regional mapping.
- **COUNTRIES (C):** Country-level identification.
- **REGIONS (R):** High-level geographic definitions.**Projection** Projection is the administrative act of selecting specific columns for display.
- **Single-Column Selection:** Retrieves a single attribute.
- **Multiple-Column Selection:** Retrieves multiple attributes, separated by commas.**The Meta-Character (*)** The asterisk (*) serves a dual purpose based on its context within the SELECT list:

1. **Meta-Character:** When used alone, it instructs the engine to retrieve all columns from the table.
2. **Multiplication Operator:** When used in mathematical expressions, it performs multiplication.**The Qualifier Rule and ORA-00923** When a developer attempts to combine the meta-character with other items (like calculated columns) without qualification, the parser fails. If the asterisk is not the first item or is not qualified, the Oracle engine misinterprets it as a multiplication operator. Consequently, it encounters a column name where it expects the FROM keyword, resulting in **ORA-00923: FROM keyword not found where expected** .To combine all columns with specific items, the * must be qualified:

- **Table Name Style:** SELECT employees.*, salary + 1000 FROM employees;
- **Table Alias Style:** SELECT e.*, salary + 1000 FROM employees e;

2. Data Manipulation: Concatenation and Quotation

Concatenation Operator (||) The developer must utilize the double pipe operator (||) to join columns or literal strings into a single character expression.

```
SELECT first_name || ' get Rs.' || salary || ' monthly salary.' FROM
employees;
```

Handling Single Quotes (The Q Operator) Literal strings in SQL are enclosed in single quotes. If the string itself contains an apostrophe (e.g., "Steven's"), the engine interprets the second quote as the end of the string, causing **ORA-01756: quoted string not properly terminated** .**The Q/Quote Operator** To bypass complex escaping, use the Q operator: Q'char...char'. This allows for custom delimiters. Note that using a space as a delimiter results in **ORA-00911: invalid character** .| Delimiter Category | Valid Delimiters | Example Syntax || ------ |

------ | ------ || **Matching Pairs** | , {}, (), <> | Q''s join date is' || **Identical Characters** | #, ^, A, 3, * | Q'#'s salary is #', Q'A's email is A' |

3. Arithmetic Operations and NULL Handling

Arithmetic Operators Calculations are performed using the four standard operators: / (Division), * (Multiplication), + (Addition), and - (Subtraction).**Operator Precedence (BFDMAS)** The Oracle engine evaluates expressions according to the **BFDMAS** hierarchy:

1. **B** - Brackets (Parentheses () have the highest priority)
2. **F** - Functions
3. **D** - Division /
4. **M** - Multiplication *
5. **A** - Addition +
6. **S** - Subtraction - (and Concatenation ||)**The DUAL Table** The DUAL table is a dummy table maintained by Oracle, containing one row and one column. It is the essential environment for testing scalar functions and arithmetic operators without querying actual business data (e.g., SELECT 45 * 67 FROM DUAL;).**NULL Logic and the NVL Function** A NULL value represents unavailable or unknown data. It is not equivalent to zero or a space.

- **The NULL Rule:** Any arithmetic calculation performed with a NULL value returns NULL.
- **NVL (Null Value Locator):** This function takes two arguments. If the first argument is NULL, it returns the second argument; otherwise, it returns the first.
- *Instruction:* Use NVL(commission_pct, 0) to ensure calculations like salary + (salary * commission_pct) do not fail for employees without commissions.

4. Database Pseudocolumns

Pseudocolumns are variables automatically managed by the database. They act like columns but are not stored in the table structure.

- **ROWID:** A unique 18-character physical address for every row. It follows the **OBFR** (Object, Block, File, Record) structure. ROWID is generated at the time of **new row insertion** . By default, rows are displayed in ascending order of their ROWID.
- **ROWNUM:** A run-time serial number generated for the rows in a result set. It is assigned in the order rows are retrieved (based on ROWID) and only for rows that satisfy the WHERE clause.

5. Sorting Data (ORDER BY)

The ORDER BY clause retrieves data in a specific sequence and must be positioned at the absolute end of the SQL statement.

- **Sort Directions:** ASC (Ascending, default) and DESC (Descending).
- **Null Handling:** Use NULLS FIRST or NULLS LAST to explicitly position empty values.
- **Sort Methods:** Data can be sorted by column name, column alias, or the numeric position of the column in the SELECT list (e.g., ORDER BY 2).

6. Top-N Analysis (OFFSET & FETCH)

Top-N analysis allows developers to retrieve a specific slice of the result set.**The FETCH Clause** Used to retrieve a specific number of rows from the top.

- **Values:** Can be an integer, formula (e.g., 4+2), or complex function (e.g., SQRT(16)+2 or POWER(36, 0.5)).
- **Decimal Logic:** Oracle uses **Floor** logic (rounding down). FETCH FIRST 7.89 ROWS ONLY will return exactly 7 rows.
- **ONLY vs. WITH TIES:** ONLY returns the exact count. WITH TIES includes additional rows if they share the same sort-key value as the last row.**The OFFSET Clause** Used to skip a set number of rows.
- *Example (Middle Analysis):* OFFSET 5 ROWS FETCH NEXT 7 ROWS ONLY skips the first 5 rows and returns the subsequent 7.

7. Data Restriction (The WHERE Clause)

The WHERE clause facilitates Selection/Restriction by applying conditions to filter rows. Note that using a column alias in a WHERE clause is forbidden and results in **ORA-00904: invalid identifier** .**Operator Categories**

- **Single-Row Operators:** =, <, >, <=, >=, <>, !=, ^=.
- **Multi-Row Operators:** Used with a list of values.
- = ANY / = SOME: Positive list matching (returns true if the value matches any item in the list).
- <> ALL: Negative list exclusion (returns true only if the value matches none of the items).
- *Logic Warning:* Developers must avoid <> ANY. Logically, <> ANY returns almost every row because any value is "not equal" to at least one item in a multi-value list, rendering the filter useless.
- **Pair-wise Comparison:** Evaluates multiple columns simultaneously. Syntax requires nested parentheses: WHERE (dept_id, mgr_id) = ANY ((50, 124), (60, 103)).**Special Operators**
- **IN:** Matches values in a set.
- **BETWEEN ... AND ...:** Matches a range (inclusive). Syntax must be BETWEEN lowervalue AND highervalue.
- **LIKE:** Pattern matching using _ (one character) and % (zero or more).
- **IS NULL / IS NOT NULL:** The only valid way to identify empty values.

8. Introduction to SQL Functions

Functions are categorized as Single-Row (Scalar) or Multi-Row.**Case Manipulation Functions**

- **UPPER / LOWER:** Standardizes case.
- **INITCAP:** Capitalizes the first letter of each word. Crucially, word separators include special characters AND **digits (numbers)** . For example, INITCAP('AMar5akbar^anTHonY') results in Amar5akbar^Anthony.**Character Manipulation (CONCAT)** The CONCAT function joins exactly two arguments. It is functionally identical to the || operator. To join three or more items, the function must be nested: CONCAT(arg1, CONCAT(arg2, arg3)).**Nested Functions** Functions can be infinitely nested.

- **Execution Flow:** Evaluated from the **innermost** function to the **outermost** .
- **Display Logic:** Only the result of the outermost function is displayed to the user; inner results serve as inputs for the next layer.